



REPRODUCTIVE HEALTH

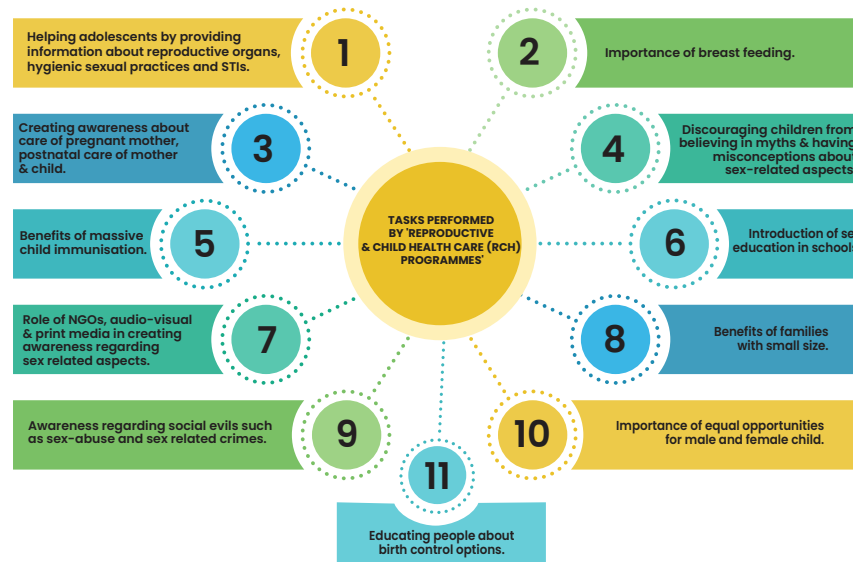
INTRODUCTION

1) DEFINITION

According to WHO, reproductive health means a total well being in all aspects of reproduction, i.e. physical, emotional, social and behavioral.

2) REPRODUCTIVE HEALTH : PROBLEMS AND STRATEGIES

India was amongst first countries in the world to initiate action plans to attain reproductive health such as Family Planning Programmes (FPP) in 1951.



3) POPULATION STABILISATION

• According to 2011 census, our population growth rate was less than 2 % i.e. 20/1000/year.

Year	World Population	Indian Population
1900	2 billion	350 million
2000	6 billion	1 billion
2011	7.2 billion	1.2 billion

• REASONS FOR INCREASE IN POPULATION SIZE :

- Decline in death rate.
- Rapid decline in maternal mortality rate (MMR).
- Decrease in infant mortality rate (IMR).
- Increase in number of people in reproductive age.
- Increase in health facilities.

- **Measures Taken By Government To Check Population Growth Rate :**
 - Motivate smaller families by using various contraceptive methods with slogans "Hum do Hamare do", advertisements and posters.
 - Urban couples adopting : "One child norm".
- **Statutory raising of marriageable age:**
 - Female to 18 years.
 - Male to 21 years.
- Incentives given to couples with small families.

4) BIRTH CONTROL/CONTRACEPTION

• Features of an ideal contraceptive:

- User-friendly.
- Easily available.
- Effective.
- Reversible.
- No/least side-effects.
- No interference with libido or act of coitus.

• There are two principle methods of birth control:

- Natural methods.
- Artificial methods.

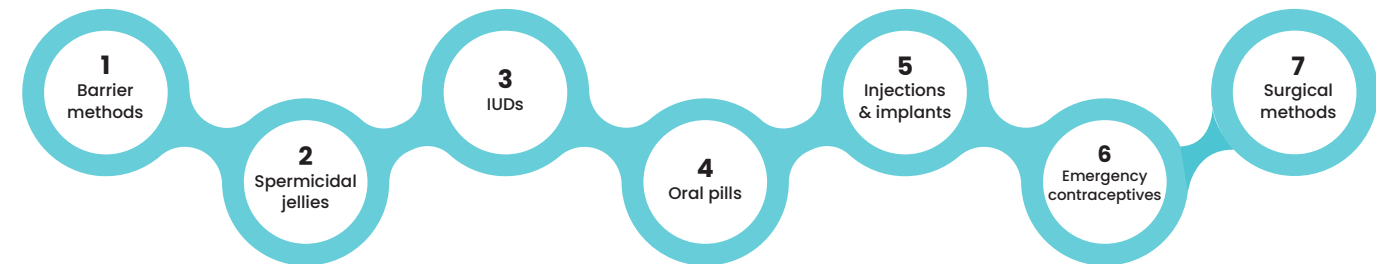
5) NATURALTRADITIONAL METHODS

- Principle of avoiding physical meeting of the egg and sperms.
- Chances of failure are high.

Method	Mode of Action (MoA)
Periodic abstinence :	Couples abstain from coitus from day 10 to 17 of the menstrual cycle i.e. fertile period.
Withdrawal method :	Insemination is avoided as the male partner, Coitus interruptus, withdraws his penis from the vagina just prior to ejaculation.
Lactational amenorrhea :	Absence of menstruation upto 6 months during period of intense lactation following parturition.

6) ARTIFICIAL METHODS

The various methods of contraception are enlisted below:



7) ARTIFICIAL METHODS

(i) Barrier methods :

- Prevent ovum and sperm from physically meeting.
- Self inserted and offer privacy to user.

(a) Condoms & its types

Made up of rubber and thin latex

Parameter	Male	Female
• Region covered	• Penis	• Vagina and cervix
• Provides protection from STIs	• Yes	• Yes
	Male condom (Nirodh)	Female condom

(b) Diaphragms, cervical caps, vaults :

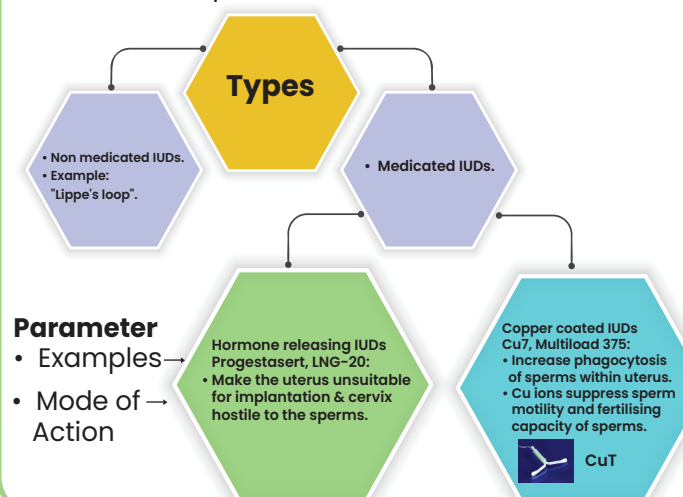
- Rubber barriers that cover the cervix during coitus.
- Reusable.
- Do not protect from STIs.
- Used by females only.

(ii) Spermicidal jellies, foams and creams :

- Kill the sperms by creating acidic pH.
- Used along with barrier methods to increase their efficiency.

(iii) Intra-uterine devices (IUDs) :

- Inserted by doctors or expert nurses in uterus.
- IUDs are one of the most widely accepted method of contraception in India.



Parameter

- Examples
- Mode of Action

(iv) Oral Contraceptive Pills (OCP,) or tablets :

Parameter	Non-steroidal	Steroidal
Example or composition.	Saheli.	Progestogens (Prg) alone or combination of Prg and Estrogens (Est).
Mode of action.	Interferes with implantation.	Inhibit ovulation & implantation; also alter the quality of cervical mucus to retard entry of sperms.
Dosage.	'Once a week' pill.	Pills have to be taken daily for a period of 21 days starting preferably within first 5 days of menstrual cycle.
Effectiveness.	High contraceptive value with very few side effects.	Pills are very effective with lesser side effects & well accepted by females.

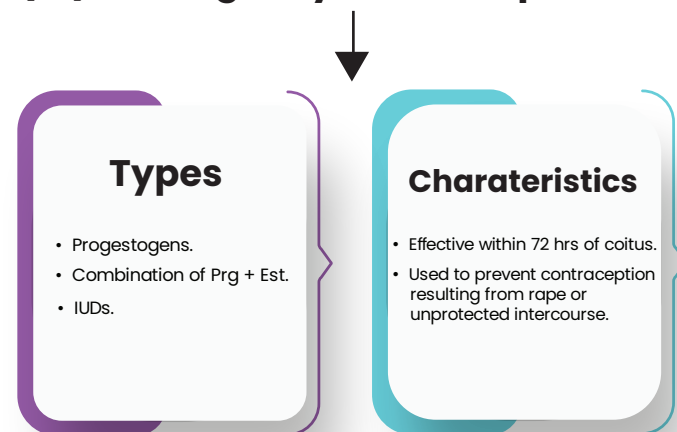
Saheli was developed at CDRI, Lucknow, Uttar Pradesh.

(v) Implants :

- Placed under skin.
- Effective periods are much longer.
- Composition: Progestagens alone/Combination of Progestagens and Estrogens.
- **Mode of Action (MoA).**
 - Inhibit ovulation and implantation.
 - Alter the quality of cervical mucus to retard entry of sperms.
 - Injections usually share similar MoA and composition as implants.



(VI) Emergency contraceptives :



8) MEDICAL TERMINATION OF PREGNANCY (MTP)/INDUCED ABORTION :

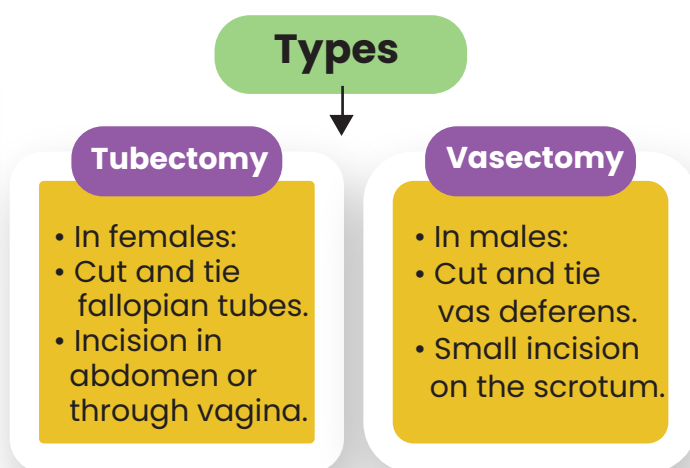
- **MTP** : Intentional or voluntary termination of pregnancy before full term.
- MTP was legalized in India in **1971**.
- **When can MTP be performed?**
- Unwanted pregnancy due to rape, failure of contraception, casual unprotected intercourse.
- If continuation of pregnancy could harm the mother of foetus or both.

• Intention behind MTP amendment act 2017 :

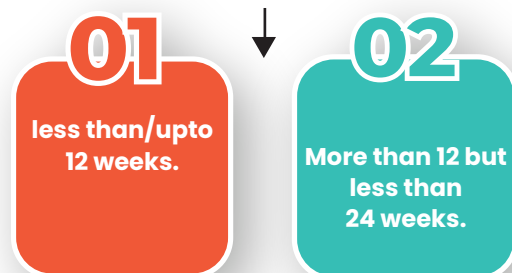
- Reduction in the incidence of illegal abortion.
- Decrease consequent maternal mortality and morbidity.
- MTPs are safe upto 12 weeks but riskier in 2nd trimester yet both are legal.
- Amniocentesis and MTPs have been misused in context of female foeticide.

(VII) Surgical/Sterilisation methods :

- Poor reversibility but highly effective.
- **Mode of action**
Blocks gamete transport.



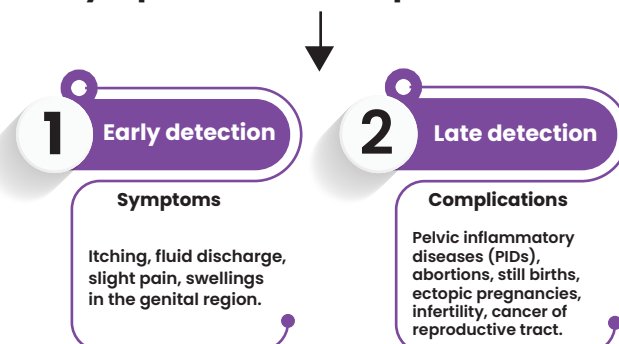
- **Permission of how many medical practitioners is needed for MTP depending on duration of gestation?**



9) AMNIOCENTESIS :

- Analyse foetal cells and dissolved substances from amniotic fluids.
- Technique used to check for genetic disorders such as Down's syndrome, hemophilia, sickle-cell anemia etc.
- Statutory ben on this technique in India to prevent female foeticide.

• Symptoms and complications of STIs :



• Preventive measures to avoid STIs:

- Avoid sex with unknown partners/multiple partners.
- Always try to use condoms during coitus.

10) SEXUALLY TRANSMITTED INFECTIONS (STIS) :

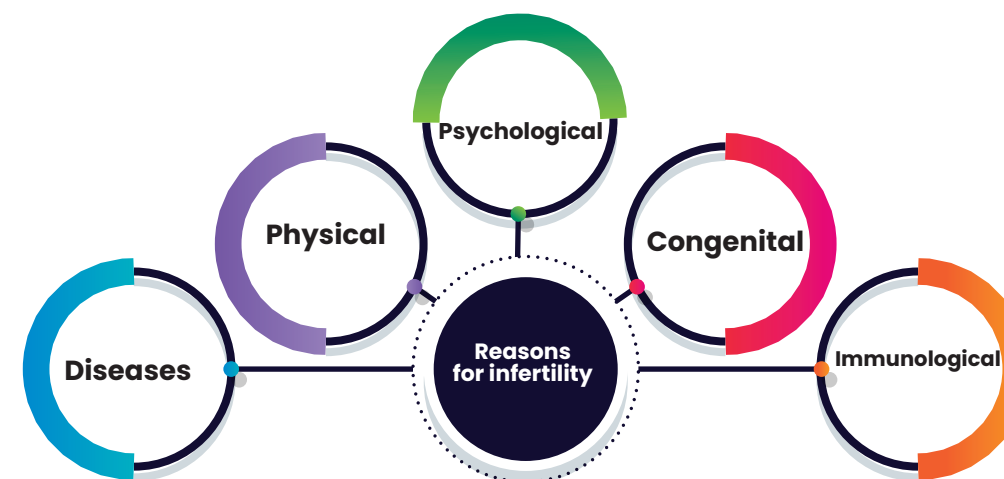
- **Alternately named:** Venereal diseases (VD) or reproductive tract infections (RTIs).
- **High vulnerability/risk group:** 15-24 years.
- **Mode of transmission (MoT):** Sexual intercourse.

Category	Disease
1 Bacterial	Gonorrhea, Syphilis, Chlamydiasis.
2 Protozoan	Trichomoniasis.
3 Viral	Genital herpes, Hepatitis-B, Genital warts, AIDS.

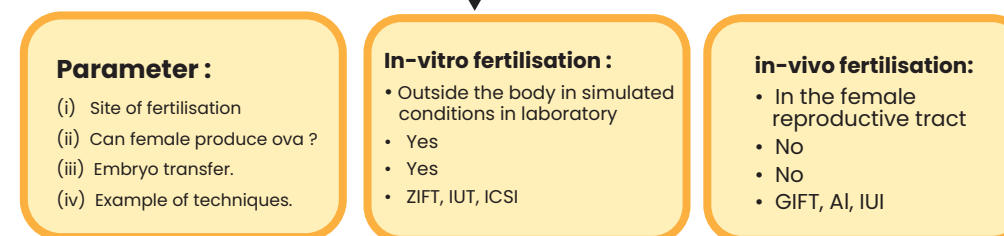
- Bacterial and protozoan diseases are completely curable if detected early and treated properly.
- **Other MoT for hepatitis-B virus and HIV infection include:**
- Sharing of injection needles, surgical instruments with infected persons.
- Transfusion of blood.
- From infected mother to foetus.

11) INFERTILITY

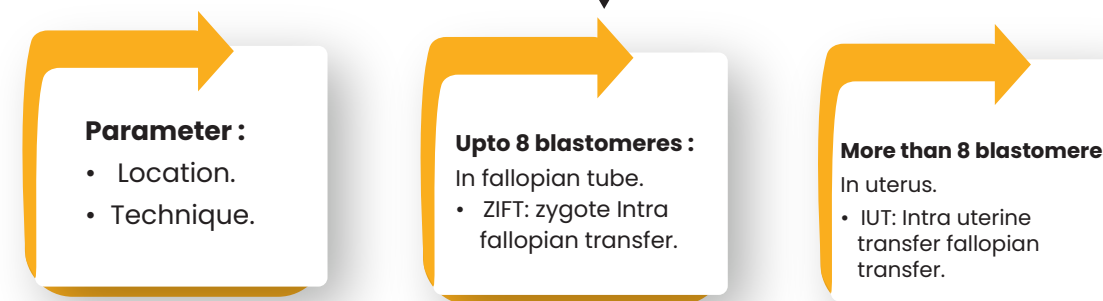
- **Infertile couple** : Unable to produce children inspite of 2 years of unprotected sexual co-habitation.



Help for infertile couples comes in the form of ASSISTED REPRODUCTIVE TECHNOLOGIES (ART).



Site of Embryo Transfer (ET) based on number of blastomeres :



• Other details of ART involved :

(i) ICSI : Intra cytoplasmic sperm injection :

- Sperm injected directly into the egg.

(ii) Artificial Insemination (AI):

- Semen introduced in vagina or uterus.
- Low sperm count or inability of male to inseminate female.
- **IUI** : Intra uterine insemination.

(iii) GIFT : Gamete intra fallopian transfer:

- Female can provide conditions for fertilisation and further development.